

Revisiting Aliasing in GAN Vocoders: Improving Intermediate Feature Representations

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Abstract—Neural audio vocoders are deep neural networks that aim to reconstruct raw audio waveforms with corresponding mel spectrograms. Despite recent advances in end-to-end TTS systems, mel spectrograms are still widely used as the intermediate output, that needs to be further processed by a vocoder, in real-world speech synthesis systems and in synthesizing other types of audio such as singing voices, music, etc. Therefore, developing fast and high-fidelity neural audio vocoders is still of great interest. Traditional transposed-convolution-based upsampling methods are known to cause aliasing artifacts. There are recent studies that incorporate anti-aliasing in the vocoder architecture. However, the upsampling layers in state-of-the-art vocoders usually remain to be primitive aliasing-inducing transposed convolutions. In this work, we study the unique frequency composition patterns of the intermediate features of the vocoder generators due to the upsampling and propose several modifications to the BigVGAN vocoder generator. More specifically, we propose to remove the anti-aliasing filters in the lower sampling rate blocks and add spectral inversion modules and frequency shuffle upsampling modules. With the modifications, we further improve the reconstruction quality of the state-of-the-art vocoder BigVGAN while achieving faster synthesis speed.

Index Terms—Neural Audio Vocoders, GAN-based Synthesis, Aliasing Artifacts, Transposed Convolution, Spectral Inversion

I. INTRODUCTION

The goal of neural audio vocoders is to reconstruct raw audio signals from audio features. Since the adaptation of WaveNet [1], neural audio synthesis has been researched profoundly and achieved high fidelity. GAN-based models achieved a balance between high-fidelity synthesis and fast generation and are the most dominant neural vocoder architecture thus far.

WaveGAN [2] was one of the first works that utilize GAN architecture for non-autoregressive adversarial audio synthesis. MelGAN [3] proposed a multi-scale discriminator (MSD) and employed a feature matching loss. Hifi-GAN [4] proposed a multi-period discriminator (MPD) and utilized mel spectrogram loss. Hifi-GAN surpasses autoregressive vocoders in terms of audio quality while achieving fast non-autoregressive generation. Univnet [5] employed location-variable convolution [6] and a multi-resolution spectrogram discriminator (MRSD). Multiband-MelGAN [7] proposed a multiband generation architecture and employed a pseudo

quadrature mirror filter bank (PQMF) for synthesizing different generated frequency bands. Avocado [8] proposed a multi-band collaborative discriminator combined with a sub-band discriminator. VNet [9] utilized full-band Mel-spectrum information as input and generates high-resolution signals through a multi-tier discriminator. To address the issue of waveform artifacts, BigVGAN [10] proposed a multi-periodic compositional structure that integrates anti-aliasing filters alongside the snake activation function [11]. Regarding discriminator architectures, BigVSAN [12] applied the slicing adversarial network (SAN) framework to vocoder design. Furthermore, FA-GAN [13] incorporated an anti-aliased dual deconvolution mechanism coupled with a fine-grained multi-resolution real-imaginary loss function. This strategy shows remarkable proficiency in mitigating high-frequency noise while better recovering spectral nuances. Gu et al. [14] constructed a novel discriminator leveraging both multi-scale time-compressed continuous wavelet transforms and multi-scale subband constant Q transforms. Collectively, these advanced discriminator strategies demonstrate remarkable efficacy in spoken and sung audio generation, yielding substantial upgrades to foundational models like HiFi-GAN and BigVGAN. GAN-based models achieved a balance between high-fidelity synthesis and fast generation and are the most dominant neural vocoder architecture thus far [15].

Transposed convolutions are widely used in generative models for upsampling. One key feature of the vanilla transposed convolution is the aliasing artifacts, as essentially, transposed convolution can be interpreted as a convolution after a zero-insertion upsampling. For image neural networks, suppressing these aliasing artifacts results in better translation-equivariant properties [16] and leads to less texture sticking and more diversity for generative models [17]. For neural audio vocoders, aliasing artifacts are known to cause artifacts such as spectral imaging [8]. Recent studies have further highlighted the importance of explicitly addressing vocoder artifacts, for example by introducing global artifact filtering mechanisms in neural vocoders [18]. Several works utilize anti-aliasing structures for the adversarial generation of audio. Alternative approaches attempt to mitigate upsampling-induced artifacts by modifying filtering strategies within GAN vocoders, such as the stacked shifted filters proposed in JenGAN [19]. For instance, Avocado [8] proposed a PQMF-based anti-aliasing discriminator architecture to help the generator learn to suppress artifacts caused by the upsampling layer. BigVGAN [10] proposed to

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add anti-aliasing filters to the activation function to suppress the aliasing caused by the non-linear activation. There are also works that utilize split-band generation, which can be seen as a form of implicit anti-aliasing, such as Multi-band MelGAN [7] and Fre-GAN2 [20].

However, despite efforts that have been made to introduce anti-aliasing in time-domain neural vocoders, completely suppressing all aliased frequencies in the intermediate layers (by introducing anti-aliasing filters after each transposed convolution layer) results in early collapse [10]. This suggests that although aliasing in neural vocoders is known to cause artifacts [21], aliasing may also be essential for generating high-fidelity audio [2].

II. PROPOSED METHOD

The overall architecture of the proposed vocoder generator is shown in Figure 1. Generally, the generator architecture follows the design in [10], which utilizes snake non-linear activation [11] and anti-aliasing filters for activation. The main difference between our proposed model and BigVGAN is the removal of anti-aliasing filters in the lower sampling rate blocks and the added frequency shuffle upsampling modules and spectral inversion modules.

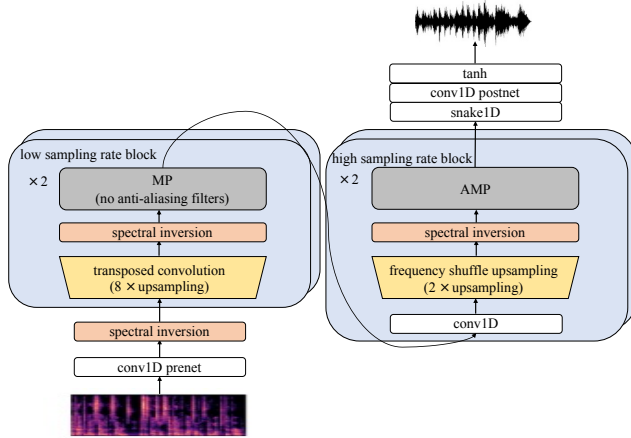


Fig. 1: The overall architecture of the generator of the proposed model.

For discriminators, we followed [10] and utilized the multi-resolution spectrogram discriminator (MSRD) [5] and the multi-period discriminator (MPD) [4]. It should be noted that our work focuses on the improvement of the generator architecture, so we chose the same discriminators as in [10].

A. Removing Filters for Lower Sampling Rate Blocks

BigVGAN utilized anti-aliasing filters for non-linear snake activations to suppress aliased high-frequencies caused by the non-linear activation. Specifically, when applying the activation, the intermediate features are first upsampled and low-passed with the windowed sinc filter, then the activation is applied, and finally, the features are low-pass filtered with the windowed sinc filter and downsampled to the original sampling rate, as shown in Figure 2.

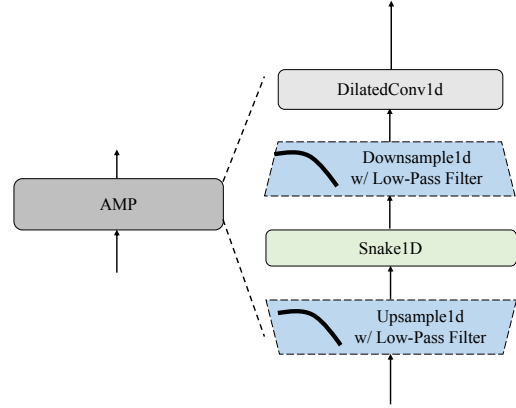


Fig. 2: The anti-aliased multi-periodicity composition (AMP) modules proposed in BigVGAN. For lower sampling rate blocks, we removed the anti-aliasing filters, which are marked with the dashed line in the figure.

Such filter mechanisms are common in non-linear DSP systems to avoid aliasing. However, as the frequency composition of features in lower sampling rate blocks of the CNN-based vocoder generator are arbitrary [10] and thus less interpretable, and aliasing can be beneficial to the overall quality [2], adding anti-aliasing for lower sampling rate blocks may not be sensible. Therefore, we propose to remove the anti-aliasing filters in the first two lower sampling rate convolution blocks, as shown in Figure 1, to reduce overhead and allow the model to be more expressive in the lower sampling rate blocks.

B. Spectral Inversion Module

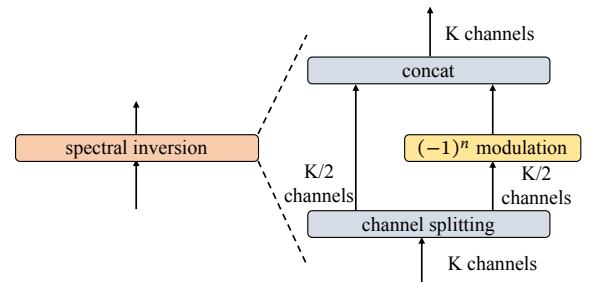


Fig. 3: Spectral inversion module. Half of the channels of the input features are multiplied with the signal $(-1)^n$.

Due to the nature of input mel spectrograms, the spectrograms of the intermediate features of the lower sampling rate blocks tend to form a stripe pattern, which provides an inductive bias. Therefore, to suppress these stripe patterns and introduce more diverse frequency components in intermediate feature maps, we propose a spectral inversion module, as shown in Figure 3. This module flips half of the feature channels frequency-wise. Spectral inversion is done by multiplying with the signal $(-1)^n$. More formally, for the n -th sample of a signal, this is done via:

$$y[n] = x[n] \cdot \cos(\pi n) = x[n] \cdot (-1)^n \quad (1)$$

When multiplying with a sinusoid with a frequency of half the sampling rate, according to the convolution theorem, the spectrogram of the modulated signal will be shifted by half the sampling frequency, leaving the negative frequencies in the original positive frequency range, thus inverting all frequencies in the original signal. This modulation technique is commonly used in digital filter design to shift prototype low-pass filters to high-pass or band-pass filters.

By inverting the frequencies of half the channels, a more diverse frequency composition is introduced in the intermediate feature maps, thus reducing bias and enabling models to generate more coherent frequency information. The stripe patterns are less obvious and the frequency composition is generally more diverse in lower sampling rate blocks, demonstrating the effectiveness of the proposed module. Apart from lower sampling rate blocks, we also introduced this module in higher sampling blocks to introduce more diverse frequencies, as this module requires negligible computational costs.

C. Frequency Shuffle Upsampling Module

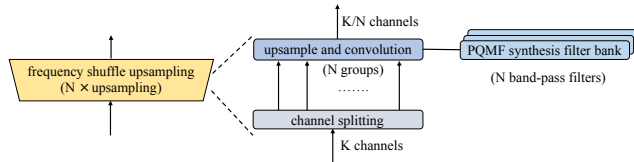


Fig. 4: frequency shuffle upsampling module using PQMF synthesis filter bank.

For the convolution blocks that operate at high sampling rates, we have found that non-linear residual blocks were ineffective in suppressing the duplicate aliased frequencies caused by the transposed convolution. This leaves the work of suppressing aliased frequencies mostly to the linear postnet convolution, which is problematic, as this linear postnet filter often leaves aliased components in the final output.

Therefore, we propose the frequency shuffle upsampling module to replace the transposed convolution layers in high sampling rate blocks to reduce the aliasing caused by upsampling in the high sampling rate blocks. It can be interpreted as a pixel shuffle upsampling operation [22] in the frequency domain. Initially, the K-channel input is split evenly into N groups along the channel dimension. The features within each group are upsampled and filtered with band-pass filters, forming the sub-bands of the upsampled signals. After filtering, band-limited features from each group are added to construct the full-band upsampled features. This upsampling process is shown in Figure 4. By allowing upsampling layers to generate different frequency bands separately, the model can be more flexible when constructing components for each frequency band.

For band-pass filters, we utilized the synthesis filter bank of the PQMF. PQMF has been used in previous neural audio

synthesis studies to synthesize mult-band output [7], [8], [23], [24]. However, to our knowledge, our work is the first to propose a structure, which serves as a direct replacement for the transposed-convolution-based upsampling layer, that employs the PQMF synthesis filter bank.

We replaced the last two $2\times$ transposed convolutions with the frequency shuffle $2\times$ upsampling modules. We do not utilize frequency shuffle for the first two lower sampling rate blocks, as the reconstruction loss for the PQMF synthesis filter bank is not negligible for such low sampling rates with 8 subbands, and empirically we found results did not improve and a significant overhead (due to the high number of channels) was added.

III. EXPERIMENTS

A. Datasets

We have used four speech datasets in our experiments, namely LJSpeech, LibriTTS, VCTK, and Aishell-3.

LJSpeech [25] The LJSpeech dataset is an English single-speaker female speech dataset recorded by Linda Johnson. The sampling rate of the audio clips from this dataset is 22050 Hz. This dataset is not pre-split. Therefore, we use the split in the official Hifi-GAN implementation repository.

LibriTTS [26] This dataset is pre-split, and the total length of the training sets adds up to 960 hours. The sampling rate of the audio clips from this dataset is 24000 Hz. It should be noted that the recording condition of some audio clips in this dataset is poor, which can pose a challenge for training a speech synthesis model on this dataset.

VCTK [27] This is a multi-speaker English speech dataset that consists of 110 speakers. Each speaker reads out about 400 sentences from a newspaper. The length of the dataset is approximately 44 hours. This dataset is only used for testing the out-of-distribution generalizing ability of the models and is not seen during training.

Aishell-3 [28] The Aishell-3 dataset is a multi-speaker Chinese speech dataset that consists of 218 speakers. The length of the dataset is approximately 85 hours. All speech data was recorded with a professional setup and underwent strict quality inspection for tone and prosody. The sampling rate of the audio clips is 44100 Hz. This dataset is only used for testing the out-of-distribution generalizing ability of the models and is not seen during training.

B. Experiment Settings

We have experimented with 2 settings, namely the band-limited setting and the universal setting. For the band-limited 80-band setting, the frequency range of the input 80-band mel spectrogram is limited to 0-8kHz and the sampling rate of the generated audio is 22.05kHz. This band-limited setting is used in [4] and is close to the traditional TTS settings. For the full-band 100-band setting, the input 100-band mel spectrogram is full-bandwidth (0-12kHz) and the sampling rate of the generated audio is 24kHz. This setting is more popular for universal vocoders and is used in [5], [10], [29]. For the 80-band setting, we use the single-speaker LJSpeech

TABLE I: Universal setting LibriTTS experiment results. $\uparrow$ and $\downarrow$ means better metrics. The best metrics are highlighted.

Model	LibriTTS				VCTK			
	MAE $\downarrow$	M-STFT $\downarrow$	PESQ $\uparrow$	UTMOS $\uparrow$	MAE $\downarrow$	M-STFT $\downarrow$	PESQ $\uparrow$	UTMOS $\uparrow$
Hifi-GAN $\dagger$	0.3848	1.3109	2.925	3.474	0.3412	1.1677	3.056	3.803
UnivNet-c32 $\ddagger$	0.2527	0.8853	3.278	3.600	0.2490	0.8508	3.329	3.884
BigVGAN-base	0.1475	0.8451	3.695	3.581	0.1403	0.8054	3.844	3.905
Proposed	0.1398	0.8433	3.768	3.561	0.1343	0.8136	3.801	3.889
	AiShell-3				LJSpeech			
	MAE $\downarrow$	M-STFT $\downarrow$	PESQ $\uparrow$	UTMOS $\uparrow$	MAE $\downarrow$	M-STFT $\downarrow$	PESQ $\uparrow$	UTMOS $\uparrow$
Hifi-GAN $\dagger$	0.3773	1.1989	2.748	2.445	0.3129	1.0013	3.161	3.966
UnivNet-c32 $\ddagger$	0.2655	0.8470	2.981	2.659	0.2825	0.9368	3.465	4.208
BigVGAN-base	0.1566	0.8178	3.570	2.545	0.1619	0.9104	3.879	4.099
Proposed	0.1478	0.8160	3.642	2.583	0.1504	0.9026	3.909	4.151

dataset and we use the same split as in the official open-source implementation of HifiGAN [4]. For the 100-band universal setting, we use the LibriTTS dataset and use the whole training set for training, which is the same as in [10]. The parameters for STFT are the same as in [10], in which the FFT size is set to 1024 and the hop size is set to 256. For the PQMF synthesis filter bank settings, we set the filter length to 256 taps, the beta to 10.0, and the cut-off ratio to 0.25. We used this same PQMF setting for both the last two high sampling rate blocks.

We chose the “BigVGAN-base” configuration of BigVGAN [10] as our main baseline model. For LJSpeech experiments, we used the 80-band BigVGAN-base configuration the hyperparameter settings for the BigVGAN baseline are not changed except for the batch size set to 16, as we found the default 32 batch size leads to unstable training on the LJSpeech dataset. This batch size is also set for our proposed model for LJSpeech experiments. For wide-band experiments, the baseline BigVGAN model is trained with the 100-band BigVGAN-base configuration in the official implementation and we did not change the hyperparameters. For our proposed model, we follow the same hyperparameter settings as BigVGAN. Both the baseline model and the proposed model are trained for 2.5 million steps. It should be noted that the log-scale snakebeta activation $f_{\alpha,\beta}(x) = x + e^{-\beta} \sin^2(e^{\alpha} x)$ is used instead of the original snake activation in the official implementation, which differs from the original BigVGAN paper.

We also tested several other pre-trained vocoders, namely MelGAN [3], Hifi-GAN [4], and UnivNet [5].

For MelGAN, we trained the model from scratch with the Hifi-GAN LJSpeech split. For Hifi-GAN, we used the “LJ_V1” checkpoint in the official implementation. For the wide-band experiments, we used the official implementation of Hifi-GAN and an unofficial implementation of UnivNet. For Hifi-GAN, we used the “UNIVERSAL_V1” checkpoint in the official implementation. The sampling rate of the output of the official Hifi-GAN implementation is 22050 Hz. Therefore, we upsampled the Hifi-GAN output to 24000 Hz to calculate metrics. For UnivNet, we used the “UnivNet-c32” checkpoint in the implementation.

C. Experiment Results

We used 4 metrics for evaluating the quality of the output samples in the experiment, namely the mean absolute error (MAE) of the mel spectrograms, multi-resolution STFT (M-STFT) distance, perceptual evaluation of speech quality (PESQ), and the UTMOS [30] predicted subjective quality scores. For the UTMOS scores, we used the pre-trained model in the official implementation. We downsampled the output to 16000 Hz when calculating the PESQ scores and the UTMOS scores to satisfy their input requirements.

For LibriTTS universal vocoder experiments, the results are shown in Table I. We can see from the table that excluding the UTMOS score metric, both our proposed model and BigVGAN significantly outperform previous models, while our proposed model outperforms BigVGAN. This shows that our proposed model improves the overall reconstruction quality of the BigVGAN universal vocoder model with large-scale training data.

One key capability of universal vocoders is the generalization ability over unseen out-of-distribution (OOD) data [10]. To verify whether this generalization ability is improved with our modifications, we tested the LibriTTS models on both Chinese speech data and unseen English speech data. The Chinese speech data we used is AiShell-3 [28]. The out-of-distribution English data is from VCTK [27] and LJSpeech [25]. The OOD results are shown in Table I. We can see that our proposed model generally outperforms the BigVGAN model, indicating our proposed modifications improve the robustness and the generalization ability of the BigVGAN base model.

We calculated the non-intrusive UTMOS [30] objective quality scores for our experiments. From the results, we can see that the UTMOS system tends to give higher scores for UnivNet. However, both in terms of other objective metrics and subjective evaluations, BigVGAN as well as our proposed model generates higher-quality samples and is less prone to artifacts than UnivNet. Moreover, the UTMOS system gives low scores for all AiShell-3-based Chinese samples, despite the generally higher recording quality of the AiShell-3 dataset. This is due to the lack of Chinese data in the UTMOS training set [31].

For band-limited LJSpeech experiments, the results are

TABLE II: Band-limited setting LJSpeech experiment results.

Model	MAE ↓	M-STFT ↓	PESQ ↑
MelGAN	0.4811	1.1669	2.143
Hifi-GAN	0.2114	0.9161	3.637
BigVGAN-base	0.1659	0.8737	3.793
Proposed	0.1613	0.8715	3.844

shown in Table II. Same as in the universal setting, our proposed model outperforms BigVGAN and previous vocoder models. This demonstrates that for applications where the training data are scarce, our proposed modifications can still improve the overall quality of the output audio samples.

The results of the band-limited and the wide-band experiment should not be compared directly, as the input mel spectrogram resolution (80 vs. 100), the sampling rate of the output audio (22050 Hz vs. 24000 Hz.), and most importantly, the training data and the scale of the training data is different in both settings. However, we can see that despite the universal setting models never being trained on the LJSpeech dataset, the PESQ scores (which are calculated at the same 16000 Hz sampling rate in both settings) on the LJSpeech dataset of the universal setting models are higher than those of the narrow-band models. This phenomenon shows the benefits of large-scale training for neural vocoders.

For subjective metrics, we conducted a 5-scale similarity mean opinion score (SMOS) experiment.

TABLE III: The synthesis speed comparison of the proposed model and the BigVGAN-base model. The unit is "× RT" represents how many times the synthesis speed is faster than the real-time.

Model	CPU ↑	GPU ↑
BigVGAN-base	0.9720	134.6
Proposed	1.3205	166.0

Moreover, we compared the synthesis speed of the proposed model and BigVGAN, as shown in Table III. We can see that with our proposed modification, the model runs faster both on CPU and GPU and even achieves faster than real-time on our tested CPU. This shows that the filters in the lower sampling rate blocks are more computationally intensive than the added proposed modules. We have only included the 80-band setting results in the synthesis speed comparison table as the only difference between the model structure in 2 settings is the linear layers in the prenet and the postnet, which constitute negligible overhead compared to the whole model.

D. Ablation Study

We also conducted an ablation study. We separately removed the frequency shuffling module and the spectral inversion module removed both modules and trained the ablation models for 2.5 million steps on the LJSpeech dataset with the band-limited setting. The results are shown in Table IV. It should be noted that the only difference between the ablation models with both the modules removed and the original BigVGAN model is the removed filters in the lower sampling rate blocks.

TABLE IV: The ablation results on LJSpeech. The speed shown in this table is GPU speed.

Model	MAE ↓	M-STFT ↓	PESQ ↑	Speed ↑
BigVGAN-base	0.1659	0.8737	3.793	134.6
w/o SI, FS	0.1619	0.8737	3.796	171.6
w/o SI	0.1615	0.8698	3.821	168.3
w/o FS	0.1635	0.8767	3.827	166.7
Proposed	0.1613	0.8715	3.844	166.0

TABLE V: AB Preference Test Results (%)

Dataset	BigVGAN	Proposed	No Preference
LJSpeech	42.3	48.7	9.0
VCTK (OOD)	28.5	61.2	10.3
Aishell-3 (OOD)	25.1	64.8	10.1

From the ablation study results, we can see that by removing anti-aliasing filters in the lower sampling rate blocks, the model does not suffer from result degradation, and the computational overhead is significantly reduced. By removing two proposed modules, we can see the results degrade from the proposed model, which shows the effectiveness of each proposed modification. Moreover, we can see that the two proposed modules are remarkably computation-friendly and only add a small amount of overhead to the overall architecture.

E. Subjective Listening Evaluation

To evaluate perceptual audio quality, we conducted AB preference and Mean Opinion Score (MOS) tests comparing the proposed vocoder to BigVGAN-base. Speech samples were selected from LJSpeech (in-domain), VCTK (English OOD), and Aishell-3 (Chinese OOD) datasets: 20 utterances each, totaling 60 paired comparisons. All used identical acoustic features. Fifteen listeners with normal hearing participated in a quiet environment via headphones.

In the AB test, listeners chose the more natural, artifact-free sample (A, B, or no preference) from randomized pairs. Results (Table V) show the proposed model preferred, especially on OOD data, indicating improved robustness.

MOS ratings (1–5 scale) were given independently to randomized samples. Results (Table VI) confirm higher scores for the proposed model, with greater gains on OOD sets, demonstrating enhanced generalization.

F. Analysis on Artifacts

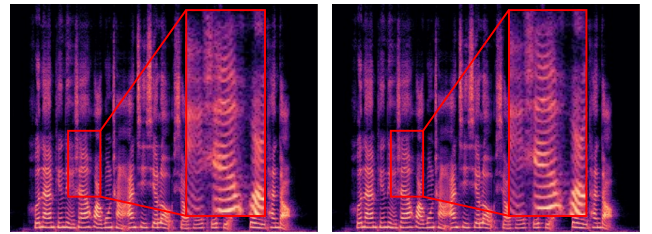


Fig. 5: The spectrograms of reconstructed audio of a Chinese speech. Left: BigVGAN. Right: Our proposed model. Note the harmonic error of the BigVGAN-generated sample. This artifact is clearly audible.

TABLE VI: MOS Results (Mean $\pm$ Std.)

Model	LJSpeech	VCTK (OOD)	Aishell-3 (OOD)
BigVGAN-base	4.21 $\pm$ 0.38	3.89 $\pm$ 0.42	3.62 $\pm$ 0.47
Proposed	4.28 $\pm$ 0.36	4.05 $\pm$ 0.39	3.91 $\pm$ 0.44

Despite the BigVGAN vocoder having already achieved high-fidelity reconstruction, there can still be errors when synthesizing OOD data. For instance, in Figure 5, we have included a comparison of the spectrograms of a Chinese speech generated by BigVGAN and our proposed model. We can see that the BigVGAN failed to generate clear and correct harmonics for this Chinese instance. As we theorize this is due to the unique frequency pattern perturbing the process of generating correct harmonics. Our proposed model can be more effective when synthesizing for OOD audio. As shown in Figure 5, our proposed model can generate correct harmonics, which further confirms the robustness of our proposed model.

IV. CONCLUSION

In this paper, we explore the aliasing caused by the upsampling in neural vocoders and show that by adding some simple modifications, namely removing the anti-aliasing filters in the lower sampling rate blocks and adding the spectral inversion modules and the frequency shuffle upsampling modules, we can improve the frequency composition of intermediate features of the current state-of-the-art GAN vocoder BigVGAN. Our experiments show that our proposed model achieves better audio quality than BigVGAN for both the traditional band-limited TTS setting and the wide-band universal setting.

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